What do you want to learn?

ALICE L WALTON SCHOOL OF MEDICINE

α–Agonists

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Adapted from: Antihistamines and Decongestants

Listen to this Brick
0:00 / 17:20 1x

Learning Objectives (4) Versions

After completing this brick, you will be able to:

1. Classify alpha agonists based on their mechanism of action.
2. Predict the effects of alpha-adrenergic agonists on various organ systems.
3. Explain the therapeutic effects of alpha-adrenergic agonists.
4. Explain the p ergic agonists and discuss strategies to minimize/prevent such effects.

CASE CONNECTION

SY, a 38-year-old male, is at your office for this six-month follow-up for his hypertension. SY's blood pressure has been elevated for the last six months and is not being controlled well with this current medication (Lisinopril/Amlodipine combination). SY states that he is regularly taking his anti-hypertensives daily, controlling his diet, and also is exercising routinely. Since SY's hypertension is not being controlled, you decide to add another agent, Clonidine, to his drug regimen. SY says that the name sounds familiar to him. " I used to take it for my ADHD (attention-deficit/hyperactivity disorder) when I was a kid. I did not know it helped my pressure", says SY.

You proceed to counsel SY on the use of the new medication and explain why you are adding this new medication. What else would you counsel SY for this medication?

Consider your answer as you read, and we'll revisit SY at the end of the brick.

GO TO CONCLUSION ↓

What Are α–Receptor Agonists and how are they classified? (LO#1)

α–Receptor Agonists

In the ANS brick, we discussed that α-adrenergic receptors are classified into α1 and α2. In this bri -receptor agonists. α1-receptors are widely distributed and are in the immediate vicinity of the sympathetic terminal. On the other hand, α2 receptors are presynaptic and regulate the release of neurotransmitters in the central and peripheral sympathetic nerve endings. In addition, recall that α1-receptors are a G_q type GPCR (Gq activation→ ↑IP3→ ↑ cytosolic Ca) whereas α2 is a G_i type of GPCR (Gi activation →↓adenylyl cyclase activation →↓ cAMP formation). As you read the brick you will see that these pathways are important in understanding the mechanism of action of the α-agonists.

ACTIVE LEARNING QUESTION

Match the GPCR to the appropriate signaling pathway

Gs	✓ Activate Adenylyl cyclase -->Increase cAMP--> Increase PKA activation
Gq	✓ Increase in IP_3 --> increase in cytosolic Ca^{2+}
Gi	✓ decrease Adenylyl cyclase activation -->Derease cAMP--> Decrease PKA activation

Explanation

Gi and Gs signal with the cAMP to PKA (protein kinase A) pathway, while Gq type GPCRs activate membrane-bound Phospholipase C to increase cytosolic (or cytoplasmic) Ca^{2+}.

How do α1–agonists work and what are they used for? (LO#2 and 3)

Sympathomimetics (mimicking the sympathetic tone) such as α-1 agonists are primarily used due to their action on the smooth muscle vasculature.

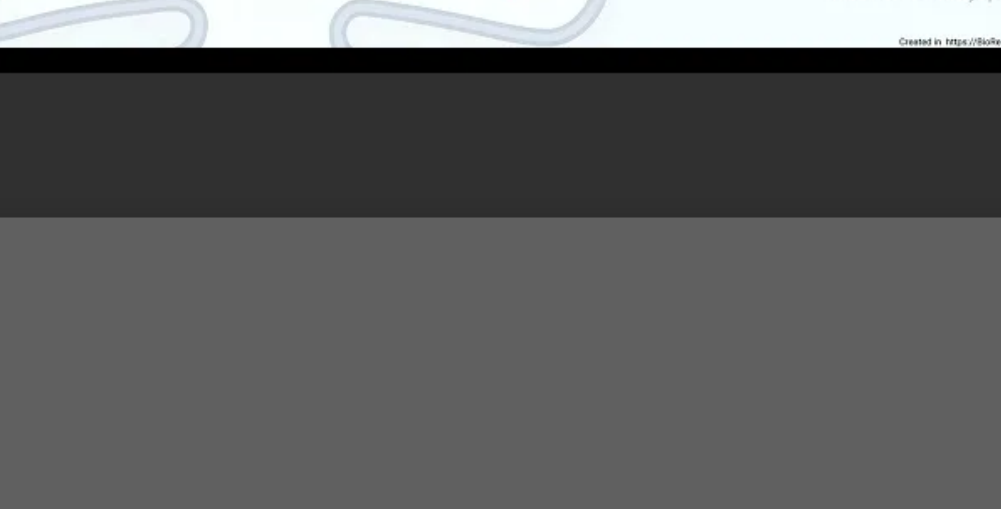
Phenylephrine

Phenylephrine is a **vasopressor** that has selective α1-agonist action.

Mechanism of action: These drugs work by binding to the α1 receptors on the smooth muscles that line the walls of blood vessels in the nose, causing them to constrict. The α1-receptors work through a G protein–coupled receptor (Gq), which activates phospholipase C (PLC). This causes conversion of phosphatidylinositol bisphosphate (PIP_2) into inositol triphosphate (IP_3) and the release of calcium from the endoplasmic reticulum and subsequent smooth muscle contraction (Figure 2). (Jordan to modify figure)

The activation of α1-induced Ca^{2+} release increases smooth muscle vasoconstriction (remember sustained cytosolic Ca^{2+} increase → muscle contraction). This increase in vasoconstriction causes a dose-dependent increase in arterial blood pressure. Thus, this mechanism is useful in acute care scenarios for patients who are very hypotensive (usually in cases of circulatory shock). Phenylephrine (as IV) is used to increase blood pressure rapidly and avoid any potential e

Mechanism of Action of α-1 adrenergic receptors:

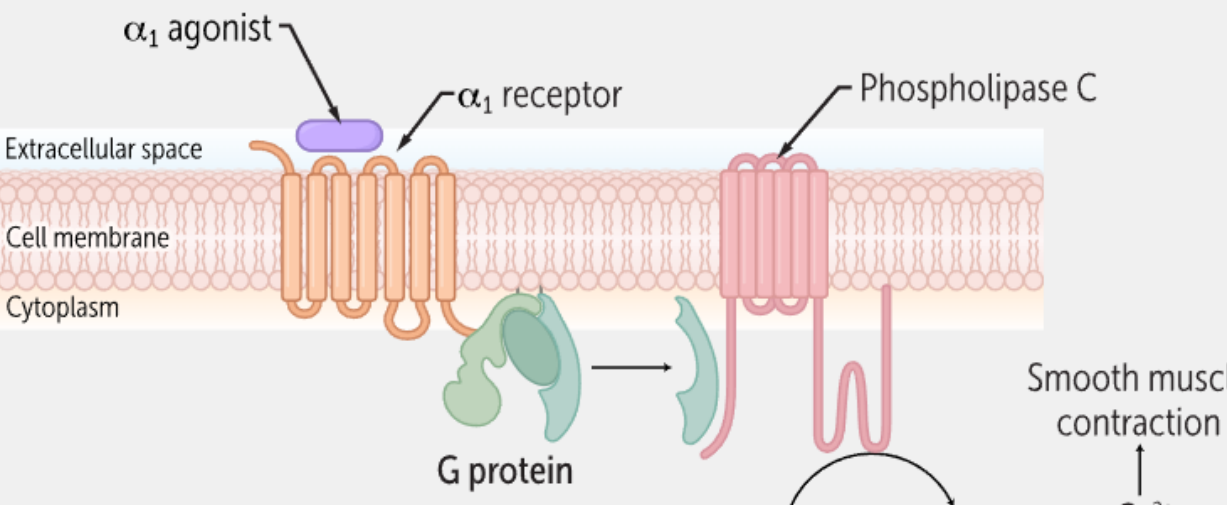

Mechanism of Action
Phenylephrine
α-1 adrenergic receptor (Gq-GPCR)
PLC
PIP2
Ca2+
ER
Smooth muscle cell cytoplasm

What receptor does phenylephrine selectively target?

Phenylephrine is also used orally and intranasally as a **nasal decongestant.** Direct stimulation of the α1-receptors on the nasal mucosa (direct effect) results in the vasoconstriction of local nasal vessels. This vasoconstrictive action decreases mucosal edema, thereby leading to a decongestant effect.

Phenylephrine stimulates alpha-adrenergic receptors within the eye (direct effect). Stimulation of α-1 receptors on the dilator muscle and arterioles of the conjunctiva in the eye causes mydriasis and vasoconstriction, respectively. They are often used as diagnostic tools in ophthalmologic investigations.

Unlike norepinephrine and epinephrine, phenylephrine is not a catecholamine. Thus, it is not metabolized by COMT(Catechol-O-methyltranferase); therefore has a longer duration of action.



QUIZ

Figure 2

Oxymetazoline

Oxymetazoline is another example of a α-1 agonist. It also causes vasoconstriction by activating α-1 receptors. The eye drops can be used to relieve minor eye irritation and facial erythema can also be treated using oxymetazoline topical cream.

CLINICAL REASONING

Norepinephrine and phenylephrine are both α1 agonists. Phenylephrine has a higher affinity to the receptor than norepinephrine. If a patient is in circulatory shock, they can have very low blood pressure (hypotensive). In such cases, vasopressors such as Norepinephrine or Phenylephrine can potentially increase BP. How and when these agents are used depends on factors such as heart rate, BP, renal function, etc.

What are the adverse effects of α1–agonists? (LO#4)

Adverse effects as a vasopressor:

- The increase in α1-receptor medicated vasoconstriction can cause **reflex bradycardia.** The increase in systemic vascular resistance (SVR) can cause a baroreceptor-mediated decrease in sympathetic outflow and, therefore, a decrease in HR.
- **Peripheral extravasation** into the surrounding tissue can lead to excessive vasoconstriction and skin necrosis.

Adverse effects as a nasal decongestant

α1-Receptor agonists have side effects because they stimulate α1-receptors outside of the nasal cavity, sometimes causing adverse effects such as hypertension (when they constrict peripheral arterioles) and anxiety (when they bind to neurons in the brain).

CLINICAL CORRELATION

It is important to note that recently an FDA Advisory Committee has determined that there is no scientific evidence supporting the effectiveness of oral phenylephrine as a nasal decongestant at the recommended dosages. The FDA will review the available evidence and the Advisory Committee's input before making any decisions regarding phenylephrine's status. Phenylephrine is found in many over-the-counter products, both as a single ingredient and in combination with others. These recommendations pertain only to orally administered phenylephrine and d

How do α2–agonists work and what are they used for? (LOs#2 and 3)

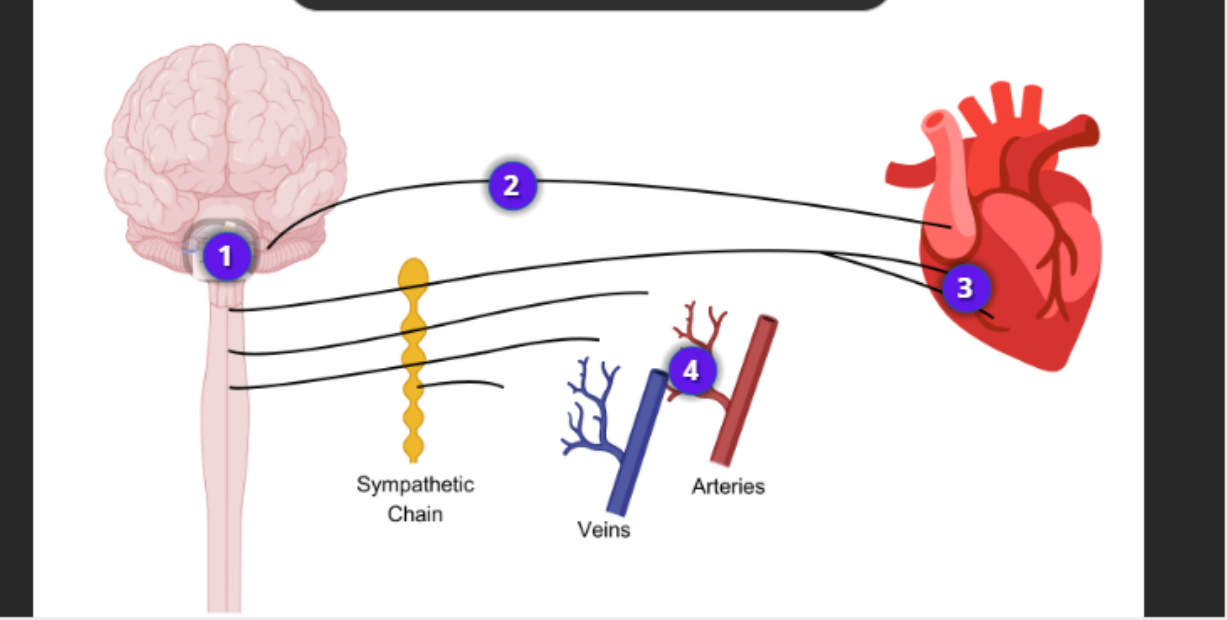
We are going to discuss two α2-agonists, clonidine and methyldopa that work with slightly different mechanisms of action on the receptors.

Clonidine

In the ANS brick, we discussed that the α_2-Receptors are found diffusely in the brain, specifically on the terminals of the presynaptic sympathetic nerves. When activated, these receptors decrease the release of norepinephrine from the neuron, reducing the activation of central nervous system (CNS) adrenergic receptors. The effects of decreased norepinephrine release in the brain results in the two important uses of clonidine:

1. As an anti-hypertensive (see Figure below)
2. In adjunct treatment for Attention-deficit/Hyperactivity disorder (ADHD)

The reduction of norepinephrine stimulation in the brain may help control unwanted behaviors associated with some psychiatric conditions (e.g., attention-deficit/hyperactivity disorder, Tourette syndrome) and symptoms of withdrawal from drugs (e.g., opioids).


Directions: Click each number to expand the mechanism of action.

What are the adverse effects of clonidine use? (LO#4)

Xerostomia (dryness of the mouth) is a common adverse effect associated with clonidine use. **Sedation** is a major adverse effect of α_2-agonists, likely resulting from reduced norepinephrine in the brain. Dizziness and fatigue can also occur with clonidine use.

CLINICAL CORRELATION

Other indications of Clonidine include attention-deficit/hyperactivity disorder and Tourette syndrome. Reducing norepinephrine stimulation in the brain may help control unwanted behaviors associated with some of these psychiatric conditions. Off-label Clonidine is used in the treatment of opioid withdrawal as an adjunct with other agents such as methadone or buprenorphine. The use of Clonidine decreases the auton sea, and anxiety.

Methyldopa

Methyldopa is another example of an α2 receptor agonist. Methyldopa is taken up into sympathetic nerve terminals and metabolized to α-methylnorepinephrine (α-MNE) which is a selective α2 receptor agonist. α-MNE is taken up into presynaptic vesicles and released in place of norepinephrine. As a result, methyldopa is often referred to as a "false neurotransmitter". Therefore, when presynaptic activation of α2 receptors in the brainstem occurs, the sympathetic outflow is decreased (decrease in available NE). This helps to lower blood pressure (see clonidine fig).

While the mechanism might indicate that methyldopa can be used in the treatment of all types of hypertensions, it is only indicated and used in the treatment of hypertension during pregnancy. It is useful to treat hypertension in pregnant patients who develop adverse effects or become intolerant to other, more commonly used medications.

CASE CONNECTION

BACK TO INTRODUCTION ↑

You proceed to counsel SY on the use of the new medication and explain why you are adding this new medication. What else would you counsel SY for this medication?

You explain to SY that the new medication, clonidine, reduces the release of certain chemicals in the brain called norepinephrine that can stop the blood pressure from increas ason why he was prescribed the same norepinephrine release can reduce the loss of focus associated with the disorder. You did counsel SY that the drug can cause dry mouth (xerostomia) and he should be staying hydrated. In addition, you state that he can get dizzy occasionally and feel tired while taking this drug. You remind SY to make sure that he books a follow-up appointment to check his blood pressure.

Summary

- α-adrenergic receptors are classified into α1 and α2. α1-receptors are widely distributed and in the immediate vicinity of the sympathetic terminal, while α2 receptors are presynaptic and regulate the release of neurotransmitters.
- α1-Receptor agonists, such as Phenylephrine, are primarily used due to their action on the smooth muscle vasculature, causing vasoconstriction and increasing arterial blood pressure.
- Phenylephrine is used as a nasal decongestant and in ophthalmologic investigations. However, the effectiveness of oral phenylephrine as a nasal decongestant is under review by the FDA.
- Adverse effects of α1-Receptor agonists such as phenylephrine include reflex bradycardia and extravasation.
- α2-Agonists, like clonidine and methyldopa, work by reducing the release of norepinephrine in the brain, thus lowering blood pressure.
- Adverse effects of α2-agonists include xerostomia, sedation, hypotension, and in elderly individuals, an increased risk for mental status changes, dizziness, and falls.

Review Questions

1. Which of the following agents is most likely to cause reflex tachycardia?

A. ○ Clonidine
B. ○ Methyldopa
C. ○ Norepinephrine
D. ● Phenylephrine

Hide Explanation

The correct answer is Phenylephrine (D). Phenylephrine increases SVR and activates the baroreceptor reflex, leading to an occurrence of reflex tachycardia. Clonidine (A) is an α-2 agonist and can decrease HR. Methyldopa (B) decreases SVR but is not likely to cause reflex tachycardia. Norepinephrine (C) increases HR and causes tachycardia but not reflex tachycardia.

2. Which mechanism explains how Clonidine lowers blood pressure?

A. ● Reduces norepinephrine release
B. ○ Increases norepinephrine release
C. ○ Increases
D. ○ Reduces phenylephrine release

Hide Explanation

The correct answer is 'Reduces norepinephrine release (A)'. Clonidine activates α2-Receptors, which decreases norepinephrine release, reducing CNS adrenergic receptor activation. 'Increases norepinephrine release (B)' is incorrect as it would raise blood pressure. 'Increases sympathetic output (C)' is incorrect as clonidine reduces it. 'Reduces phenylephrine release' is incorrect (D), Phenylephrine is not an endogenous substance.

3. Which adverse effect is commonly associated with clonidine use?

A. ○ Anxiety
B. ○ Salivation
C. ○ Tachycardia
D. ● Xerostomia

Hide Explanation

The correct answer is xerostomia (A). Xerostomia, or dryness of the mouth, is a common adverse effe wn adverse effect. Tachycardia (C) is not a listed known effect of clonidine.

4. Which of the following agents is recommended for treating hypertension during pregnancy?

A. ○ Clonidine
B. ● Methyldopa
C. ○ Oxymetazoline
D. ○ Phenylephrine

Hide Explanation

The correct answer is Methyldopa (B). Methyldopa is commonly used in the treatment of hypertension during pregnancy, especially when the patient is unresponsive to other medications. (A) Clonidine, α2 agonist, is not the most commonly recommended agent for treatment. Oxymetazoline (C) and Phenylephrine (D) are α1 agonists that increase blood pressure and cause hypertension.

How would you rate this Brick?

unresponsive to other medications. (A) Clonidine, α2 agonist, is not the most commonly recommended agent for treatment. Oxymetazoline (C) and Phenylephrine (D) are α1 agonists that increase blood pressure and cause hypertension.

How would you rate this Brick?

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Classify alpha agonists based on their mechanism of action.~~
02 ☑ ~~Predict the effects of alpha-adrenergic agonists on various organ systems.~~
03 ☑ ~~Explain the therapeutic effects of alpha-adrenergic agonists.~~
04 ☑ ~~Explain the potential adverse effects of alpha-adrenergic agonists and discuss strategies to minimize/prevent such effects.~~

Objective Confidence: 4/4